



BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Lab manual

CSIS Dept, WILP

Link to Lab portal



Cluster Lab



Console Lab



Virtual Lab



Remote Lab



Link for the lab through eLearn portal

The screenshot shows a web browser window with the URL `elearn.bits-pilani.ac.in`. The page header includes the BITS Pilani logo and the text "WORK INTEGRATED LEARNING PROGRAMMES". In the top right corner, there is a user profile for "Santosh Bangaru". The main content area features a background image of a woman studying in a library. Overlaid on this image is the text "eLearn Explore Courses & Live Lectures" followed by "Get instant access to course details and live lectures with expert instructors." Below this text is a blue button labeled "Access Courses & Live Lectures >". A large red arrow points from the text "Click on this button" to this button. A "Steps Recorder" window is visible in the top right corner of the browser, indicating that the screen is being recorded.

Steps Recorder - Recording Now

This tool is being phased out. [Click here](#) to explore alternatives.

Pause Record Stop and Review Settings

Santosh Bangaru

eLearn
Explore Courses & Live Lectures
Get instant access to course details and live lectures with expert instructors.

Access Courses & Live Lectures >

Click on this button

<https://elearn.bits-pilani.ac.in/academics/>

Link for the lab through eLearn portal

The screenshot shows a web browser window with the URL `elearn.bits-pilani.ac.in/academics/`. The page header includes the BITS Pilani logo and the text "WORK INTEGRATED LEARNING PROGRAMMES". A navigation bar contains links for "Home", "My Academics" (which is highlighted), and "WILP Policies". The main content area features a large banner with the title "My Academics" and a description: "The My Academics page is your central hub for all learning activities, featuring easy access to My Courses, eLibrary, My Virtual Labs, and Viva/Project." Below the banner, there is a row of buttons: "My Virtual Labs" (which is highlighted with a red rectangle and a red arrow pointing to it from the text "Click on this button"), "Viva / Project", and "Academics Feedback >". A "Steps Recorder" window is visible in the top right corner, indicating that the page is being recorded.

My Academics

The My Academics page is your central hub for all learning activities, featuring easy access to My Courses, eLibrary, My Virtual Labs, and Viva/Project.

Click on this button

My Virtual Labs

Viva / Project

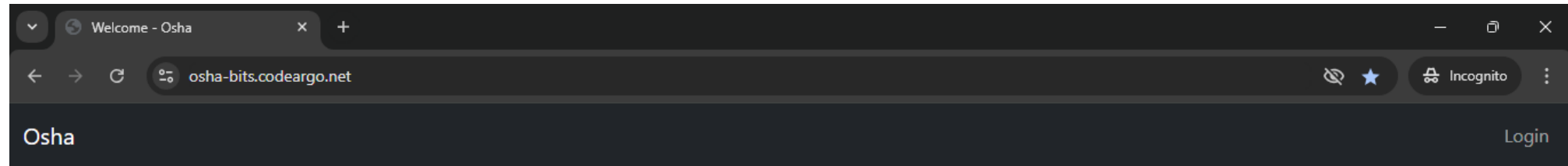
Academics Feedback >

Link for the lab through eLearn portal

The screenshot shows a web browser window with the URL `elearn.bits-pilani.ac.in/academics/#myVirtualLabs`. The page has a navigation bar with links for Home, My Academics (active), WILP Policies, My Courses, eLibrary, My Virtual Labs (active), and Viva / Project. There is also an Academics Feedback link. The main heading is 'My Virtual Labs', followed by a description: 'My Virtual Labs offers students access to a wide range of virtual experiments across various courses, providing hands-on learning opportunities in a simulated environment to enhance their academic experience.'

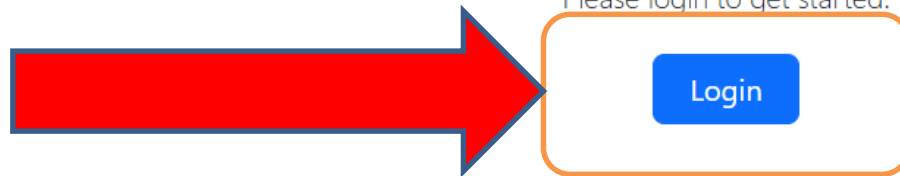
Below the description, there are two cards. The first card is titled 'CSIS - Connect to Virtual Lab' and has a blue button labeled 'View >'. A large red arrow points to this button with the text 'Click on this button'. The second card is titled 'CSIS Lab - User Manual' and also has a blue button labeled 'View >'. A yellow notification banner at the top right states: 'Steps Recorder - Recording Now. This tool is being phased out. Click here to explore alternatives.' with buttons for Pause Record, Stop and Review, and Settings.

Osha Portal Dashboard



Welcome to Lab Access Portal

Please login to get started.



Enter credentials(Same as eLearn Credentials)

My Academics - eLearn Integrated Portals Login - eLearn Integrated Portals

idp.bits-pilani.ac.in/idp/Authn/UserPassword

WORK INTEGRATED LEARNING PROGRAMMES

BITS Email Address

santosh.bangaru@hyderabad.bits-pilani.ac.in

Password

Remember Me [Forgot password?](#)

Log >

[Click here for login instructions](#)

An institution deemed to be a University estd. vide Sec.3 of the UGC Act, 1956 under notification #F.12-23/63.U-2 of June 18, 1964

innovate achieve lead

Steps Recorder - Recording Now

This tool is being phased out. [Click here](#) to explore alternatives.

Pause Record Stop and Review Settings

You have now successfully logged in to CSIS lab portal.

Lab Access Booking - Osha

o sha - bits . codeargo . net / booking /

W School

Osha

WILP CSIS Lab

Course / Experiment:

Date:

Slot:

Book

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0e4aebdc0646e60e9	12m 59s	21h 47m	saved	Start
Big Data Systems (Cluster Lab) (Node-1)	i-0ec0e002c81695ddc	10m 12s	21h 49m	saved	Start
Big Data Systems (Cluster Lab) (Node-2)	i-0e73fc5ce538079ce	11m 1s	21h 48m	saved	Start

Select the type of lab and click on the Icon



Link to Lab portal



Cluster Lab



Console Lab



Virtual Lab



Remote Lab



Refer to the Course/Experiment dropdown in the lab portal to know type of lab:

Course / Experiment:

--- Select Course / Experiment ---

--- Select Course / Experiment ---

S1-25-NSP-Sys Prog

Video Analytics (Remote Lab)

S1-25-NSP-Sys Prog

Cluster Lab

This provides direct access to cluster of virtual machines.
All the software tools required for the course is installed on these machines.



Select the course and click on “Launch” button to launch the Cluster machines

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

--- Select Course / Experiment ---

- APACHE HADOOP CLUSTER (OPENSTACK)
- APACHE KAFKA (OPENSTACK)
- APACHE SPARK (OPENSTACK)
- APACHE STORM (OPENSTACK)
- COCKROACHDB CLUSTER (OPENSTACK)
- MONGODB CLUSTER (OPENSTACK)
- MPI CLUSTER (OPENSTACK)
- NFS CLIENT-SERVER (OPENSTACK)
- POSTGRESCL CLUSTER (OPENSTACK)
- Testing tools-Appsv1
- DOCKER (OPENSTACK)

Machine	Start Time	End Time	Slot Counted
APACHE HADOOP	Nov. 18, 2024, 8 a.m.	Nov. 18, 2024, 10 a.m.	✓
APACHE HADOOP	Nov. 21, 2024, 10 a.m.	Nov. 21, 2024, noon	✓
APACHE HADOOP	Nov. 25, 2024, 10 a.m.	Nov. 25, 2024, noon	✓
APACHE HADOOP	Nov. 27, 2024, 10 a.m.	Nov. 27, 2024, noon	✓
DOCKER	Nov. 27, 2024, 2 p.m.	Nov. 27, 2024, 3 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 6, 2024, 8 p.m.	Dec. 6, 2024, 10 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 14, 2024, noon	Dec. 14, 2024, 2 p.m.	✓

Lab Access Booking

Course / Experiment: Applied Machine Learning_SESAP Date: --- N/A

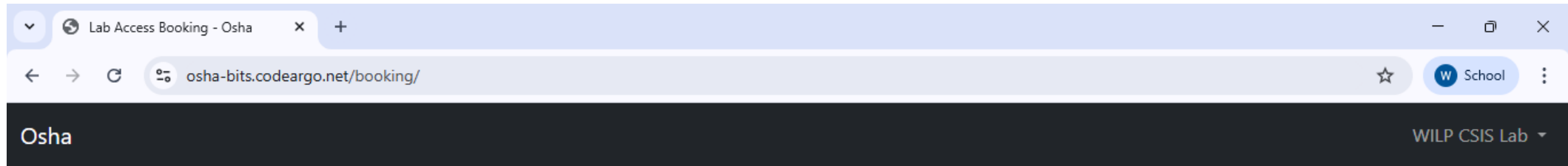
Click on “Launch”

Launch

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

After clicking launch, you can now see the cluster machines as shown below.
Click on “Start” button to start the machines



Course / Experiment:

Date:

Slot:

--- Select Course / Experiment ---

--- Select Date ---

--- Select Slot ---

Book

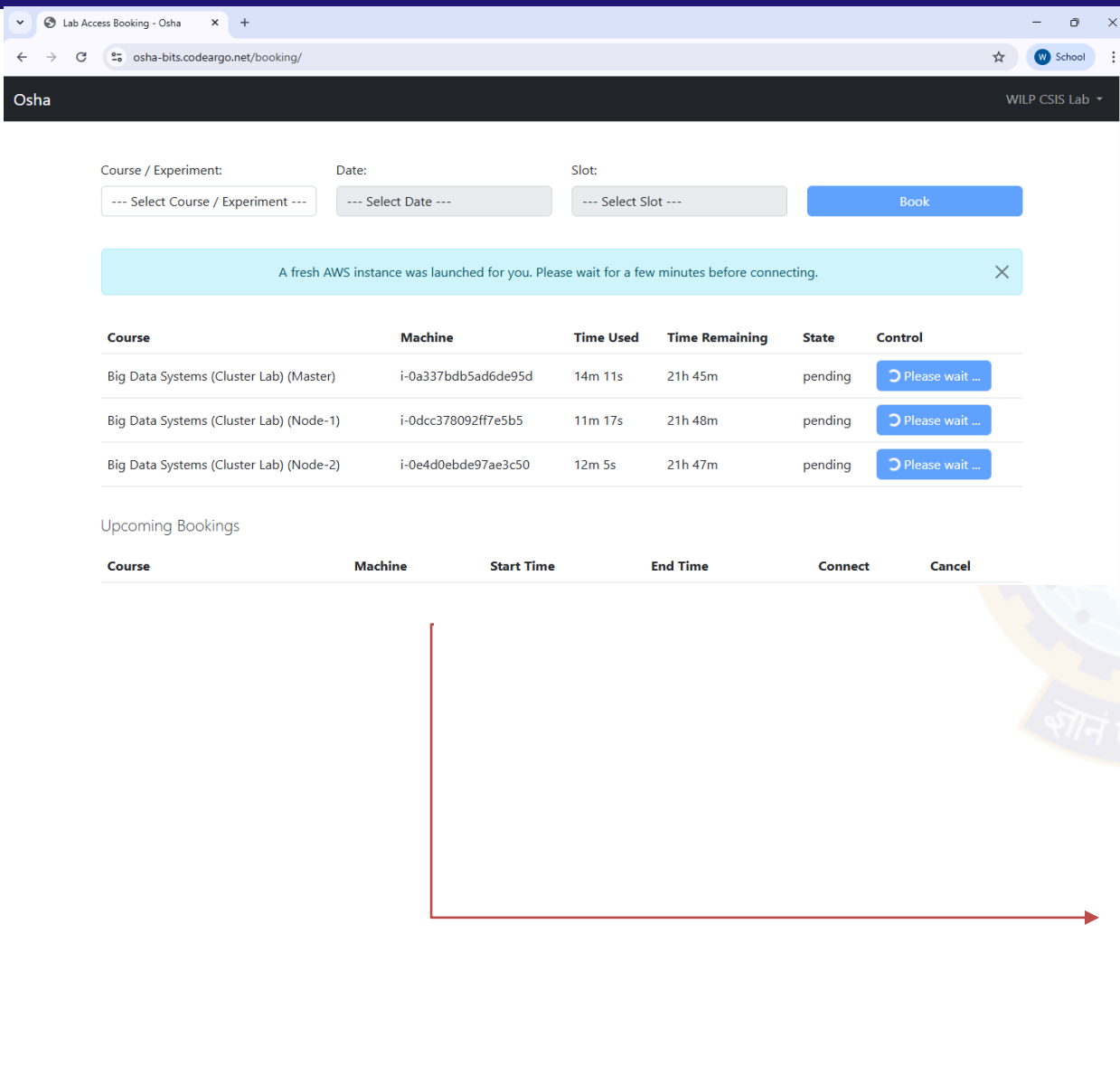
Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0e4aebdc0646e60e9	12m 59s	21h 4		Start
Big Data Systems (Cluster Lab) (Node-1)	i-0ec0e002c81695ddc	10m 12s	21h 49m	saved	Start
Big Data Systems (Cluster Lab) (Node-2)	i-0e73fc5ce538079ce	11m 1s	21h 48		Start

Click here

Click here

Click here

After clicking on “Start” button, wait for the machines to be launched.

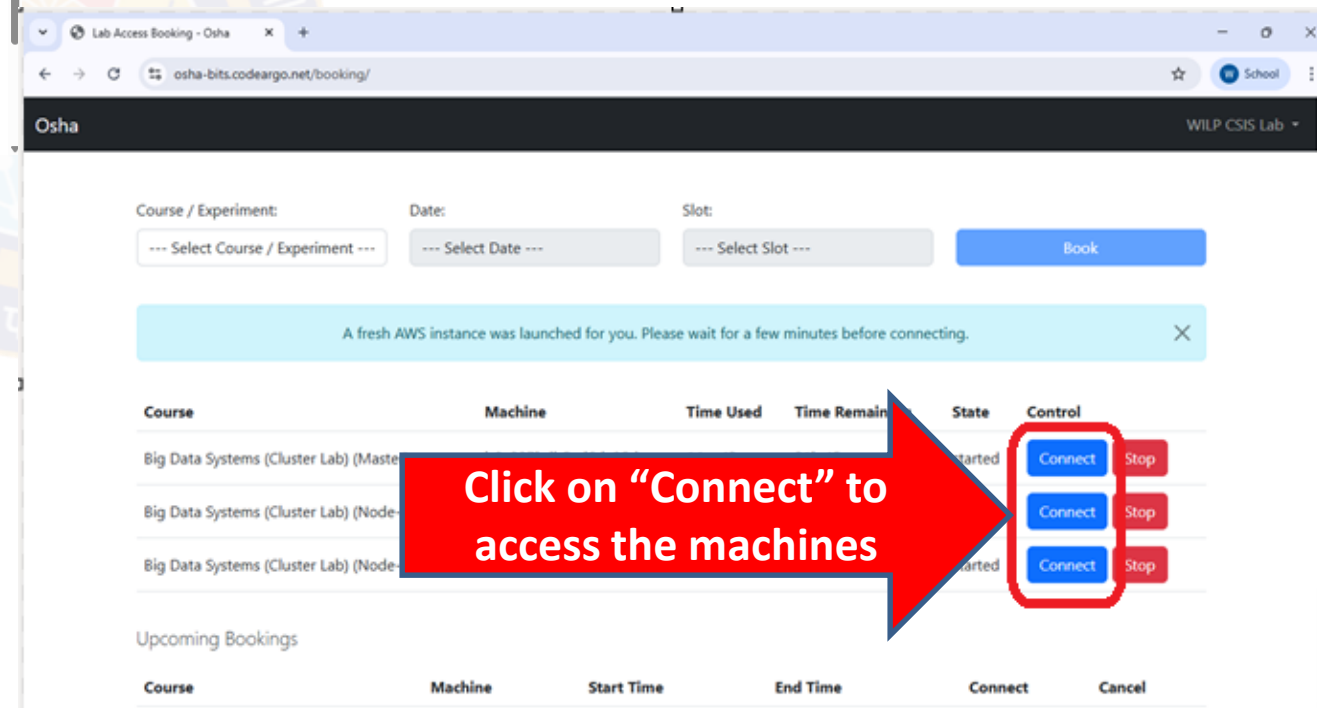


The screenshot shows the Osha Lab Access Booking interface. At the top, there are dropdown menus for "Course / Experiment:", "Date:", and "Slot:", followed by a "Book" button. Below this, a light blue notification bar states: "A fresh AWS instance was launched for you. Please wait for a few minutes before connecting." Below the notification is a table with the following data:

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	14m 11s	21h 45m	pending	Please wait ...
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	11m 17s	21h 48m	pending	Please wait ...
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 5s	21h 47m	pending	Please wait ...

Below the table, there is a section titled "Upcoming Bookings" with a table that has columns: "Course", "Machine", "Start Time", "End Time", "Connect", and "Cancel". A red arrow points from the "Upcoming Bookings" section to the right-hand screenshot.

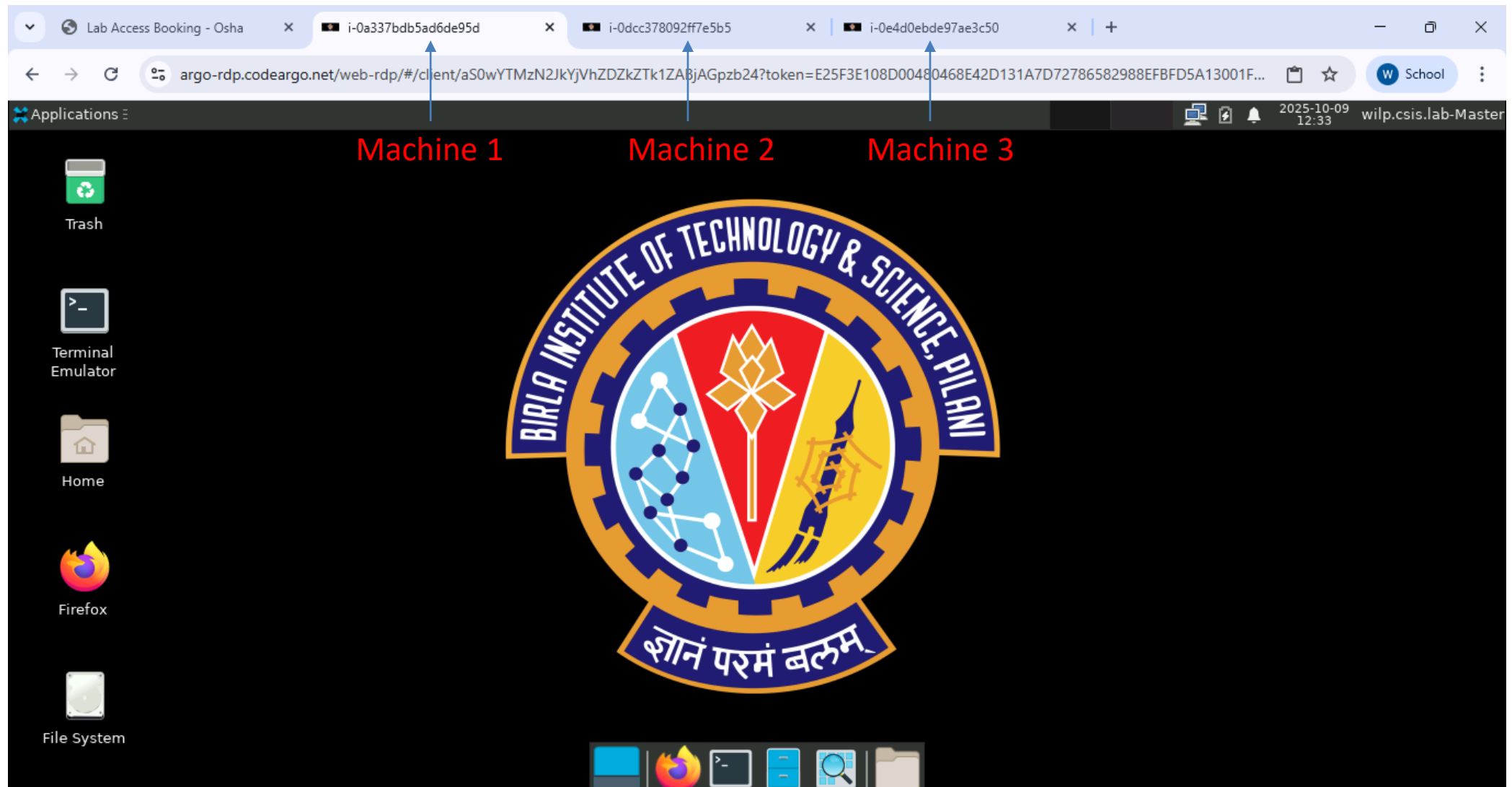
Please note that, cluster lab does not implement “idle timeout” and automatically stop the machines. You need to manually stop all the machines else ***machines will run out of allotted time.***



The screenshot shows the Osha Lab Access Booking interface, similar to the first one, but with the "State" column now showing "started" for all three machines. The "Control" column now contains "Connect" and "Stop" buttons for each machine. A large red arrow points from the text "Click on 'Connect' to access the machines" to the "Connect" buttons in the table.

Click on “Connect” to access the machines

Lab machine



Stop all the lab machines once your lab work is completed.

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha WILP CSIS Lab

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a...				Connect
Big Data Systems (Cluster Lab) (Node-1)	i-0d...				Connect
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 36s	21h 47m	started	Connect

Upcoming Bookings

Course	Machine	Start Time	End Time	Connect	Cancel
--------	---------	------------	----------	---------	--------

Click on "Stop"

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha WILP CSIS Lab

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- Book

Course	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	20m	21h	stopping	Please wait ...
Big Data Systems (Cluster Lab) (Node-1)	i-0d...			stopping	Please wait ...
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	12m 36s	21h 47m	stopping	Please wait ...

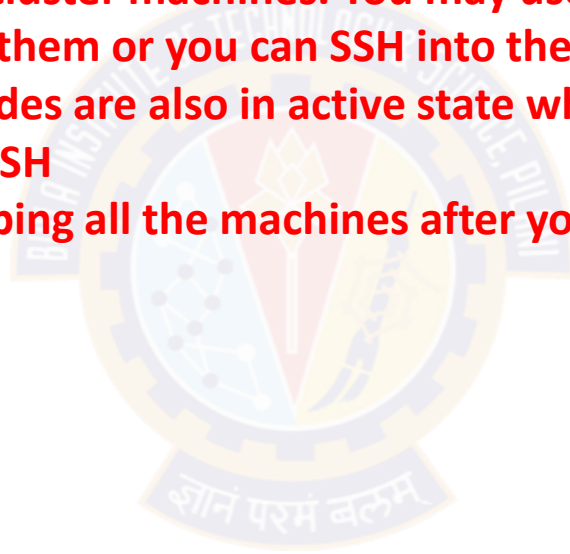
Upcoming Bookings

Course	Machine	Start Time	End Time	Connect	Cancel
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	Connect	Cancel

State is now "Stopping"

Important instructions

- Follow the instructions share by lab instructor/ labsheet to make each machine accessible with one another
- SSH access is enabled in all the cluster machines. You may use the worker nodes by directly connecting to them or you can SSH into them through Master Node. Make sure the worker nodes are also in active state while you are trying to accessing them through the SSH
- Please ensure that you are stopping all the machines after your work is completed



Console Lab

This provides direct access to Cloud console, where you will get access to different services which are enabled for the course



Select the course and click on “Launch” to launch the AWS Console

Lab TA details for HCL Courses | CSIS Virtual Cloud Lab Request | Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- [Book](#)

- Select Course / Experiment ---
- APACHE HADOOP CLUSTER (OPENSTACK)
- APACHE KAFKA (OPENSTACK)
- APACHE SPARK (OPENSTACK)
- APACHE STORM (OPENSTACK)
- COCKROACHDB CLUSTER (OPENSTACK)
- MONGODB CLUSTER (OPENSTACK)
- MPI CLUSTER (OPENSTACK)
- NFS CLIENT-SERVER (OPENSTACK)
- POSTGRESQL CLUSTER (OPENSTACK)
- Testing tools-Appsv1
- DOCKER (OPENSTACK)

Machine	Start Time	End Time	Slot Counted
APACHE HADOOP	Nov. 18, 2024, 8 a.m.	Nov. 18, 2024, 10 a.m.	✓
APACHE HADOOP	Nov. 21, 2024, 10 a.m.	Nov. 21, 2024, noon	✓
APACHE HADOOP	Nov. 25, 2024, 10 a.m.	Nov. 25, 2024, noon	✓
APACHE HADOOP	Nov. 27, 2024, 10 a.m.	Nov. 27, 2024, noon	✓
DOCKER	Nov. 27, 2024, 2 p.m.	Nov. 27, 2024, 3 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 6, 2024, 8 p.m.	Dec. 6, 2024, 10 p.m.	✓
APACHE HADOOP CLUSTER (OPENSTACK)	Dec. 14, 2024, noon	Dec. 14, 2024, 2 p.m.	✓

Inbox (123) - santosh.bangaru@ | Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

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Osha Report CSV Upload Administration Santosh Bangaru

Lab Access Booking

Course / Experiment: Applied Machine Learning_SESAP Date: --- N/A [Launch](#)

Click on “Launch”

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Click on “Launch” to access the Cloud Console

o sha-bits.codeargo.net/booking/

Osha Report CSV Upload Administration Santosh Bangaru

Course / Experiment: Date: Slot:

--- Select Course / Experiment --- --- Select Date --- --- Select Slot --- Book

Course	Credits ⓘ	State	Control
S1-25-NSPCC-CC	\$0.00 / \$10.00 (0.00%)		Launch
Cloud Computing Testing (console lab)	\$0.00 / \$2.00 (0.00%)	success	Connect

Click on Launch

Course	Machine	Time Used	Time Remaining	State	Control
S2-24-NSP2-AIML Robo - Ubuntu	i-072add1e62a260b1d	35m 38s	21h 24m	saved	Start

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Click on Connect to access the Cloud Console page

osha-bits.codeargo.net/booking/

Osha Report CSV Upload Administration Santosh Bangaru

Course / Experiment: Date: Slot: Book

--- Select Course / Experiment --- --- Select Date --- --- Select Slot ---

Course	Credits ⓘ	State	Control
S1-25-NSPCC-CC	\$0.00 / \$10.00 (0.00%)	expired	Launch
Cloud Computing Testing (console lab)	\$0.00 / \$2.00 (0.00%)		Connect

Click on Connect

Course	Machine	Time Used	Time Remaining	State	Control
S2-24-NSP2-AIML Robo - Ubuntu	i-072add1e62a260b1d	35m 38s	21h 24m	saved	Start

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ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1

Search [Alt+S]

Asia Pacific (Mumbai) Account ID: 6304-9302-9178 santosh.bangaru/santosh.bangaru-cloud-computing...

Console Home Info

Reset to default layout + Add widgets

Recently visited Info

No recently visited services

Explore one of these commonly visited AWS services.

EC2 S3 Aurora and RDS Lambda

View all services

Applications (0) Info

Create application

Region: Asia Pacific (Mumbai)

Select Region ap-south-1 (Current Region) Find applications

No applications

Get started by creating an application.

Create application

Go to myApplications

CloudShell Feedback

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Signing out from the Console

ap-south-1.console.aws.amazon.com/console/home?region=ap-south-1

aws Search [Alt+S] Asia Pacific (Mumbai) Account ID: 6304-9302-9178 santosh.bangaru/santosh.bangaru-cloud-computing...

Console Home [Info](#)

Recently visited [Info](#)

No recently visited services

Explore one of these commonly visited AWS services.

[EC2](#) [S3](#) [Aurora and RDS](#) [Lambda](#)

[View all services](#)

Applications (0) [Info](#)

Region: Asia Pacific (Mumbai)

Select Region
ap-south-1 (Current Region) [Find ap...](#)

Name	Description
No applicat...	

[Go to myApplications](#)

Account ID
6304-9302-9178

Account color
Unset

Federated user
santosh.bangaru/santosh.bangaru-cloud-computing-session

[Account](#)
[Organization](#)
[Service Quotas](#)
[Billing and Cost Management](#)

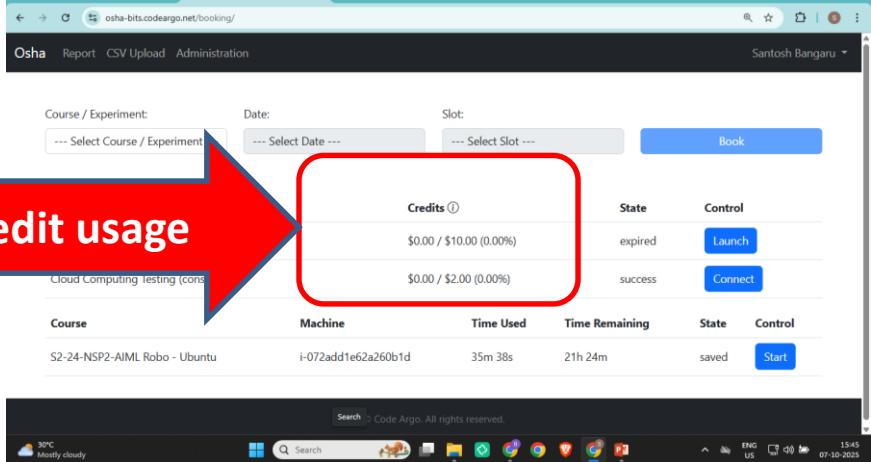
[Turn on multi-session support](#)

Click here to Signout [Sign out](#)

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Instructions to follow:

- Please ensure that all services you used are properly terminated once your work is completed.
 - Lab credits will continue to be consumed if services are left running. (Note: Make sure to terminate the EC2 instances instead of merely stopping them, since the attached EBS volumes will continue to incur charges even in stopped state.)
 - The AWS console session remains active for one hour after you log in. To continue, simply click the Launch button again in the OSHA portal.
 - For Cloud Computing course, only the pre-defined instance are allowed to be launched.
 - "t3a.micro",
 - "t3a.small",
 - "t3a.medium",
 - "t3a.nano"
 - No extra credits shall be provided
 - On 100% usage of given credits, account shall be locked down permanently and you are no longer will be able to access the AWS console. All the services consumed shall be wiped out.
 - Regularly monitor your email for credit usage alerts. You can also check your consumed credits in the OSHA portal under your booking, as shown in the screenshot below.
- (Note: AWS updates credit usage within 24–48 hours of consumption.)



The screenshot displays the OSHA portal interface. A red arrow labeled "Credit usage" points to a table showing credit consumption. Below this, a table lists active instances with their machine IDs, time used, time remaining, and state.

Credits ⓘ	State	Control
\$0.00 / \$10.00 (0.00%)	expired	Launch
\$0.00 / \$2.00 (0.00%)	success	Connect

Course	Machine	Time Used	Time Remaining	State	Control
S2-24-NSP2-AIML Robo - Ubuntu	i-072add1e62a260b1d	35m 38s	21h 24m	saved	Start



Virtual Lab

This provides direct access to a single virtual machine.
All the software tools required for the course is installed in the machine.



Select the course and click on “launch” button.

Course / Experiment: Applied Machine Learning_SESAP

Date: --- N/A ---

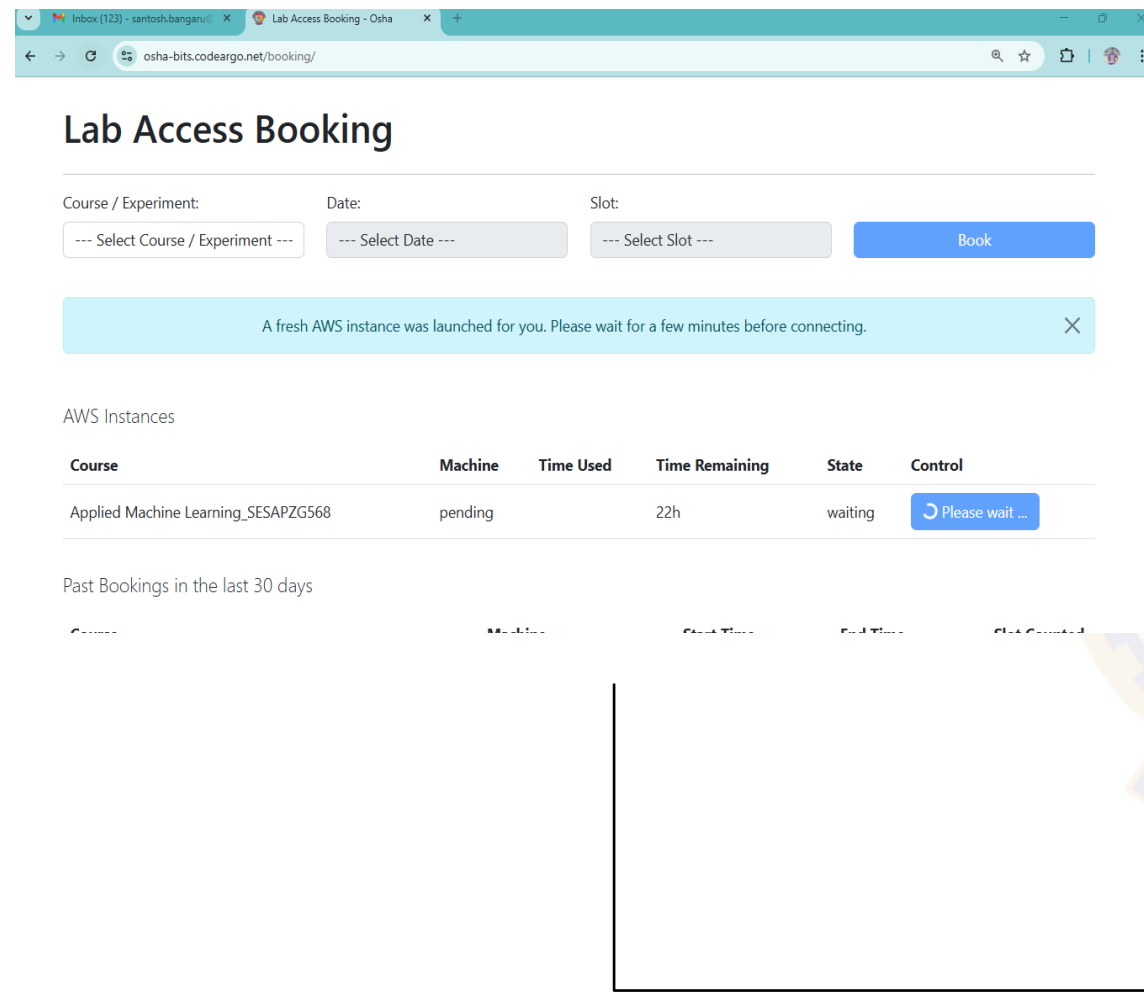
Click on “Launch”

Launch

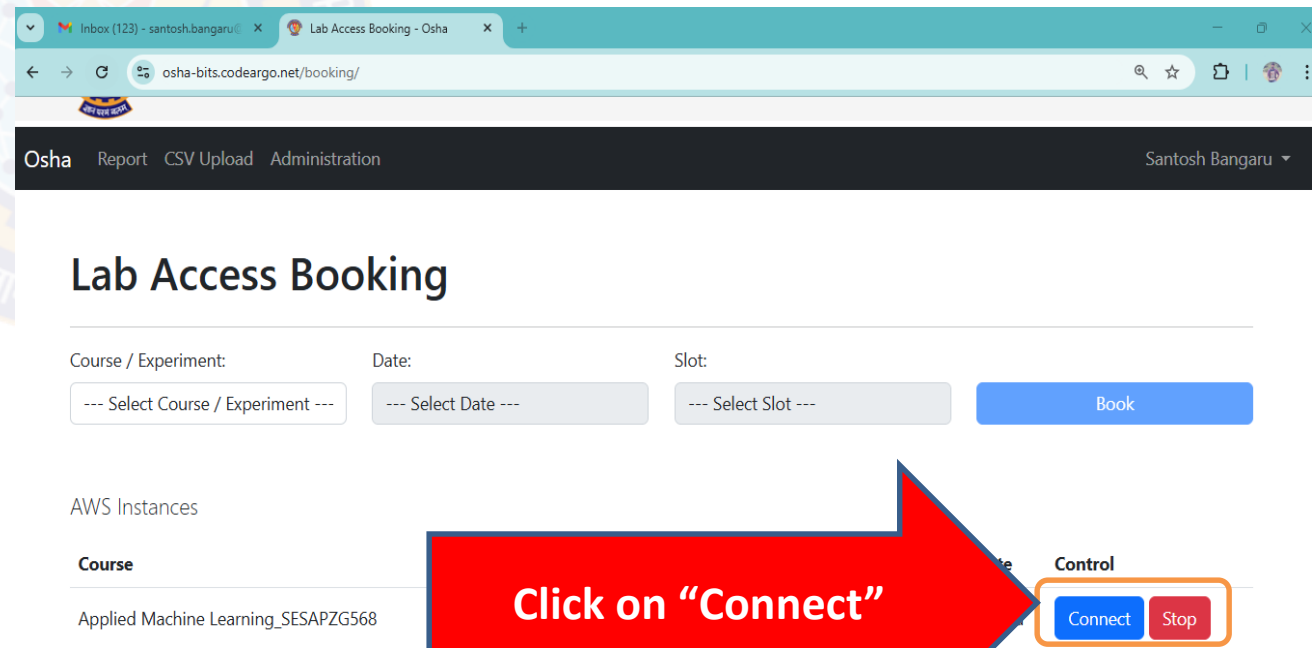
Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Upon launching the machine, Click on “Connect” to access the lab



The screenshot shows the 'Lab Access Booking' page. At the top, there are three dropdown menus for 'Course / Experiment', 'Date', and 'Slot', each with a placeholder '--- Select ... ---'. To the right of these is a blue 'Book' button. Below the dropdowns is a light blue notification box that says 'A fresh AWS instance was launched for you. Please wait for a few minutes before connecting.' with a close button (X). Under the heading 'AWS Instances', there is a table with columns: Course, Machine, Time Used, Time Remaining, State, and Control. The table contains one row for 'Applied Machine Learning_SESAPZG568' with a 'pending' machine, '22h' time remaining, and a 'waiting' state. The 'Control' column for this row has a blue button that says 'Please wait ...'. Below the table, there is a section for 'Past Bookings in the last 30 days' with a table that has columns for Course, Machine, Start Time, End Time, and Slot Counted, but it is mostly obscured.



This screenshot shows the 'Lab Access Booking' page after the machine has been launched. The 'Book' button is still present. The 'AWS Instances' table now shows the machine state as 'waiting' instead of 'pending'. The 'Control' column for the 'Applied Machine Learning_SESAPZG568' row now contains two buttons: a blue 'Connect' button and a red 'Stop' button. A large red arrow points from the text 'Click on “Connect”' to the 'Connect' button. The page also features a navigation bar at the top with links for 'Osha', 'Report', 'CSV Upload', and 'Administration', and a user profile 'Santosh Bangaru' on the right.

Click on “Web Console” to access the Machine

Inbox (123) - santosh.bangaru@

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

Report

CSV Upload

Administration

Santosh Bangaru

Lab Access Bookin

Course / Experiment:
--- Select Course / Experiment ---

Book

Connect

Please use the following credentials once you've launched the connection:

Username: ccloud
Password: ccloud

Click here

Web Console

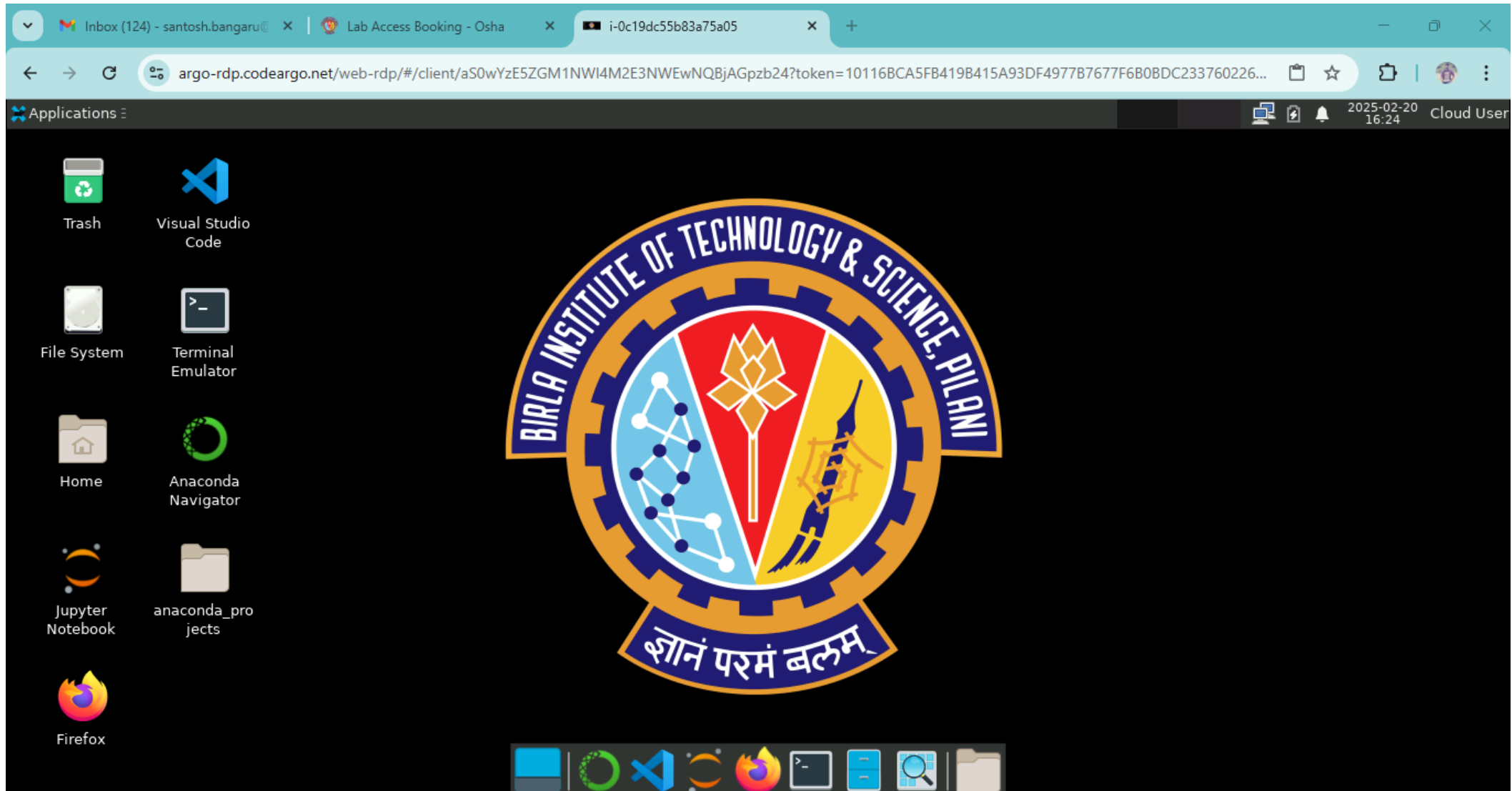
Note the username and password as you will need them when the machine gets locked

AWS Instances

Course	Machine	Time Used	Time Remaining	State	Control
Applied Machine Learning_SESAPZG568	i-0c19dc55b83a75a05	20m 50s	21h 39m	started	Connect Stop

Past Bookings in the last 30 days

You are now successfully logged into the Lab Machine



STOP the machine

osha-bits.codeargo.net/booking/

Lab Access Booking

Course / Experiment: --- Select Course / Experiment --- Date: --- Select Date --- Slot: --- Select Slot --- [Book](#)

AWS Instances

Course	Machine	Time	Control
Applied Machine Learning_SESAPZG568	i-0c19dc55b83a75a05	32	Stop

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 10 a.m.	Feb. 6, 2025, noon	✓
Big Data Analytics (S2-24_BAZG525) - Apache Framework	APACHE FRAMEWORK - 1	Feb. 6, 2025, 4 p.m.	Feb. 6, 2025, 6 p.m.	✓

Click here to STOP
the machine

Note: Kindly STOP the machine upon completing your work. The lab hours shall be consumed automatically if you forget to STOP the lab. No extra lab hours shall be provided.



Remote Lab

This provides direct access to GPU based container in a Private Cloud @Hyderabad Campus
Environment: Kubernetes Application: Kubeflow



Select the course and click on “Book” button to book your lab machine.

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment:
--- Select Course / Experiment ---
--- Select Course / Experiment ---
Big Data Systems (Cluster Lab)
Cloud Computing (Console Lab)
Introduction to DevOps (Virtual Lab)
Video Analytics (Remote Lab)
Big Data Systems (Cluster Lab) (Node-1)
Big Data Systems (Cluster Lab) (Node-2)

Date:
--- Select Date ---

Slot:
--- Select Slot ---

Machine	Time Used	Time Remaining	State	Control
i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	Start
i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	Start
i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	Start

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

Select Course, Date and slot

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

Osha

WILP CSIS Lab

Course / Experiment:
Video Analytics (Remote Lab)

Date:
--- Select Date ---

Slot:
--- Select Slot ---

Course (Virtual Lab)	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	Start
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	Start
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	Start

Past Bookings in the last 30 days

Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

Click here

Book

You have now booked your slot and you can see it under “Upcoming bookings” section

Lab Access Booking - Osha

osha-bits.codeargo.net/booking/

School

Course (Virtual Lab)	Machine	Time Used	Time Remaining	State	Control
Big Data Systems (Cluster Lab) (Master)	i-0a337bdb5ad6de95d	22m 47s	21h 37m	saved	<button>Start</button>
Big Data Systems (Cluster Lab) (Node-1)	i-0dcc378092ff7e5b5	19m 33s	21h 40m	saved	<button>Start</button>
Big Data Systems (Cluster Lab) (Node-2)	i-0e4d0ebde97ae3c50	20m 21s	21h 39m	saved	<button>Start</button>

Upcoming Bookings

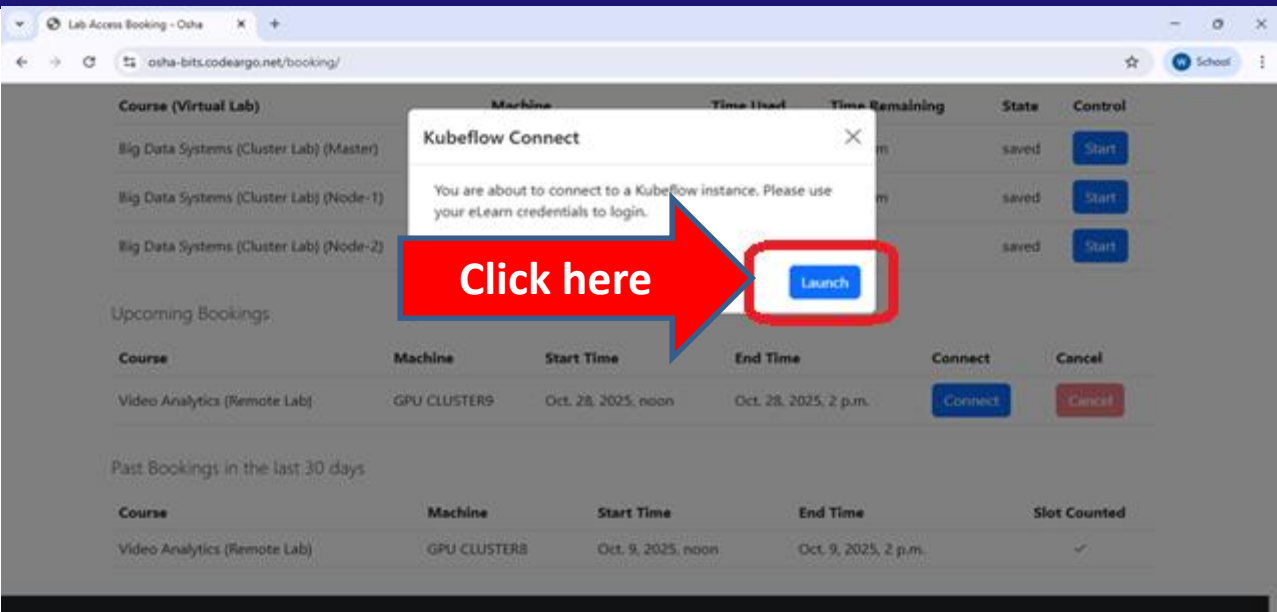
Course	Machine	Start Time	End Time	Connect	Cancel
Video Analytics (Remote Lab)	GPU CLUSTER9			<button>Connect</button>	<button>Cancel</button>

Past Bookings in the last 30 days

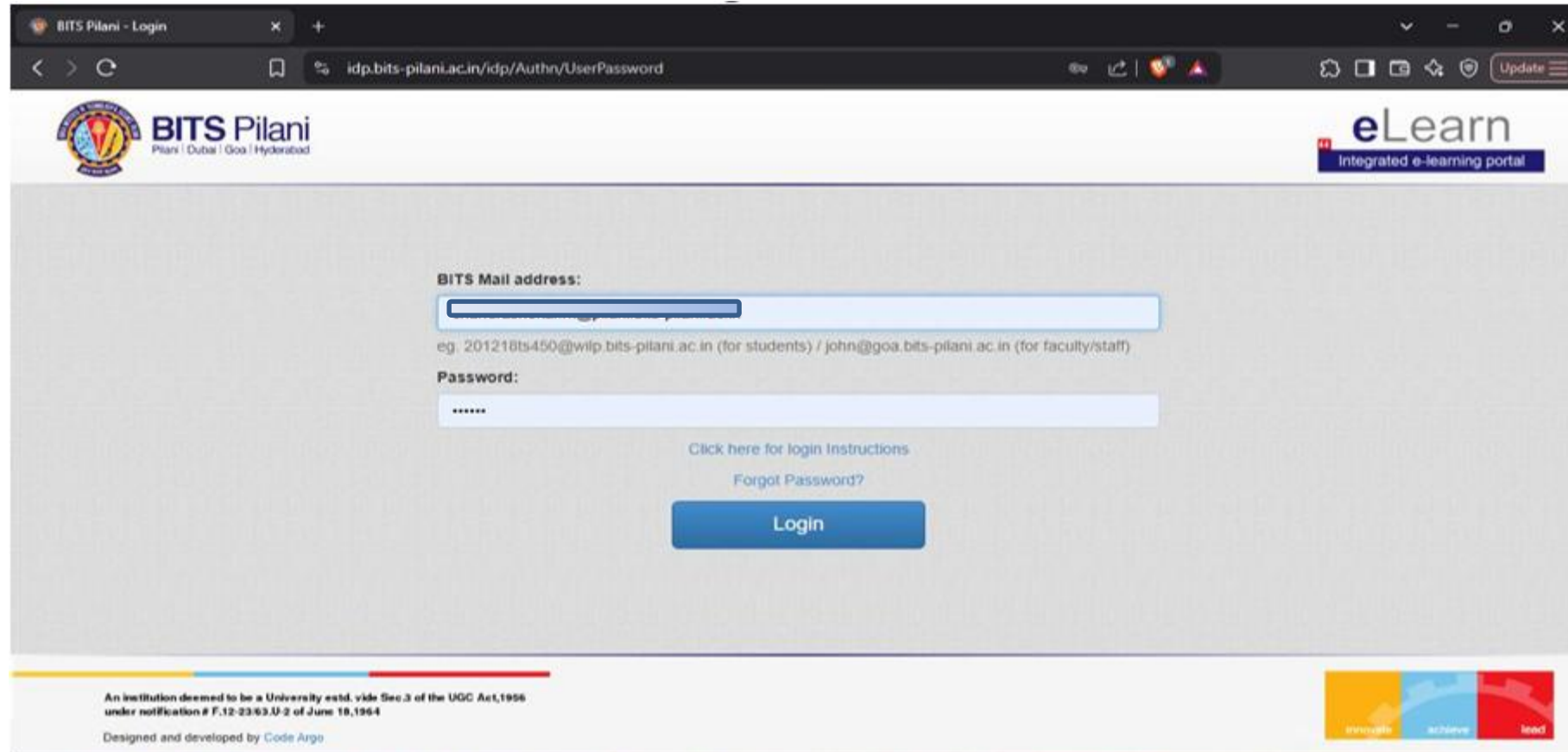
Course	Machine	Start Time	End Time	Slot Counted
Video Analytics (Remote Lab)	GPU CLUSTER8	Oct. 9, 2025, noon	Oct. 9, 2025, 2 p.m.	✓

Click here to access the lab

Click on the “Launch” button and Click on “Login with BITS Pilani SSO”



Enter eLearn portal's user credentials and Login



The screenshot shows a web browser window with the address bar displaying "idp.bits-pilani.ac.in/idp/Authn/UserPassword". The page features the BITS Pilani logo on the left and the eLearn logo on the right. The main content area contains a login form with two input fields: "BITS Mail address:" and "Password:". Below the email field, there is a text example: "eg. 201218ts450@wip.bits-pilani.ac.in (for students) / john@goa.bits-pilani.ac.in (for faculty/staff)". Below the password field, there are two links: "Click here for login Instructions" and "Forgot Password?". A blue "Login" button is positioned below these links. At the bottom of the page, there is a footer with text on the left and a graphic on the right. The text on the left states: "An institution deemed to be a University estd. vide Sec.3 of the UGC Act,1956 under notification # F.12-23/63.U-2 of June 18,1964" and "Designed and developed by Code Argo". The graphic on the right consists of three colored squares (orange, blue, red) with the words "innovate", "achieve", and "lead" respectively.

BITS Pilani
Pilani | Dubai | Goa | Hyderabad

eLearn
Integrated e-learning portal

BITS Mail address:

eg. 201218ts450@wip.bits-pilani.ac.in (for students) / john@goa.bits-pilani.ac.in (for faculty/staff)

Password:

[Click here for login Instructions](#)

[Forgot Password?](#)

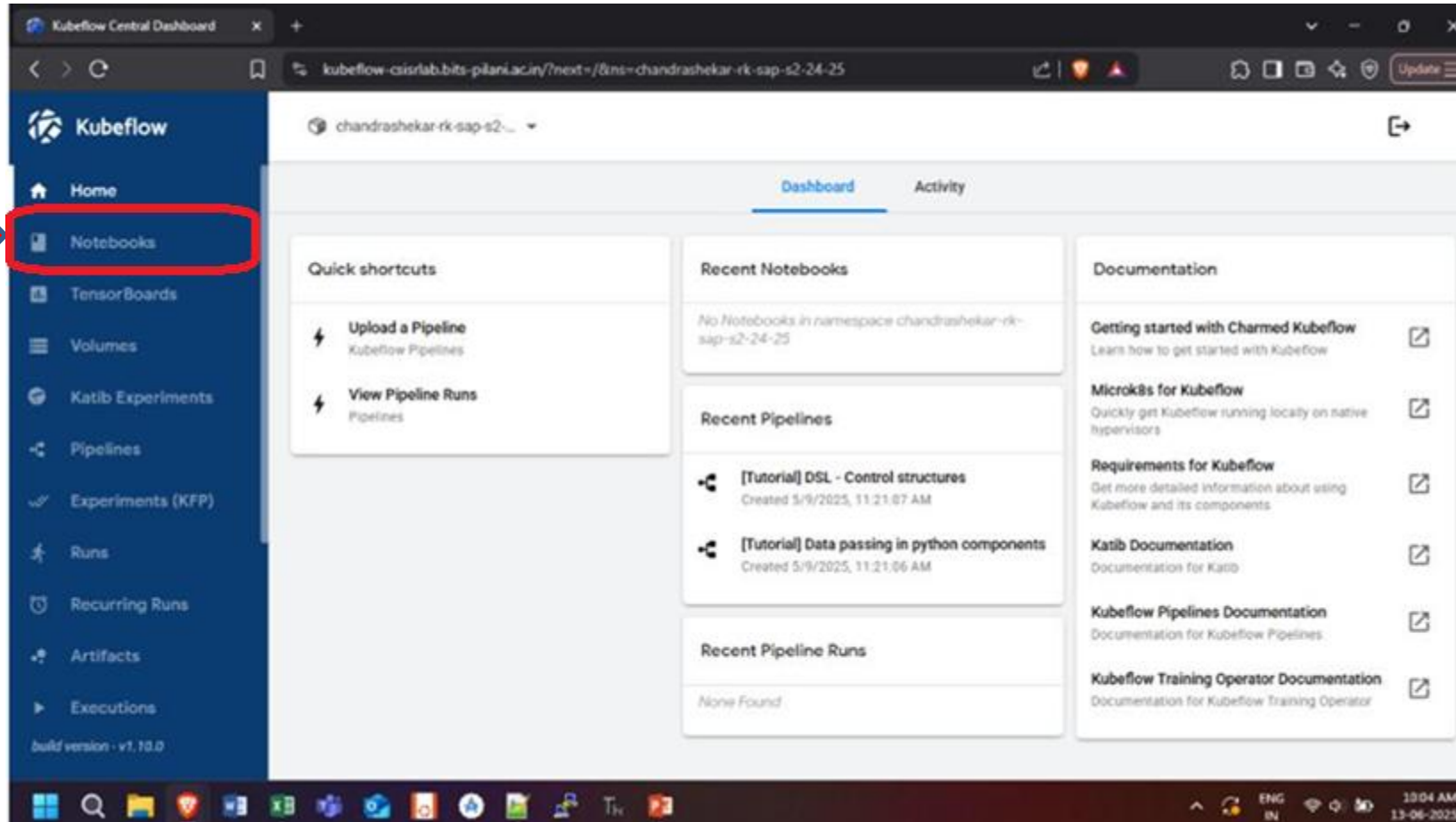
Login

An institution deemed to be a University estd. vide Sec.3 of the UGC Act,1956
under notification # F.12-23/63.U-2 of June 18,1964

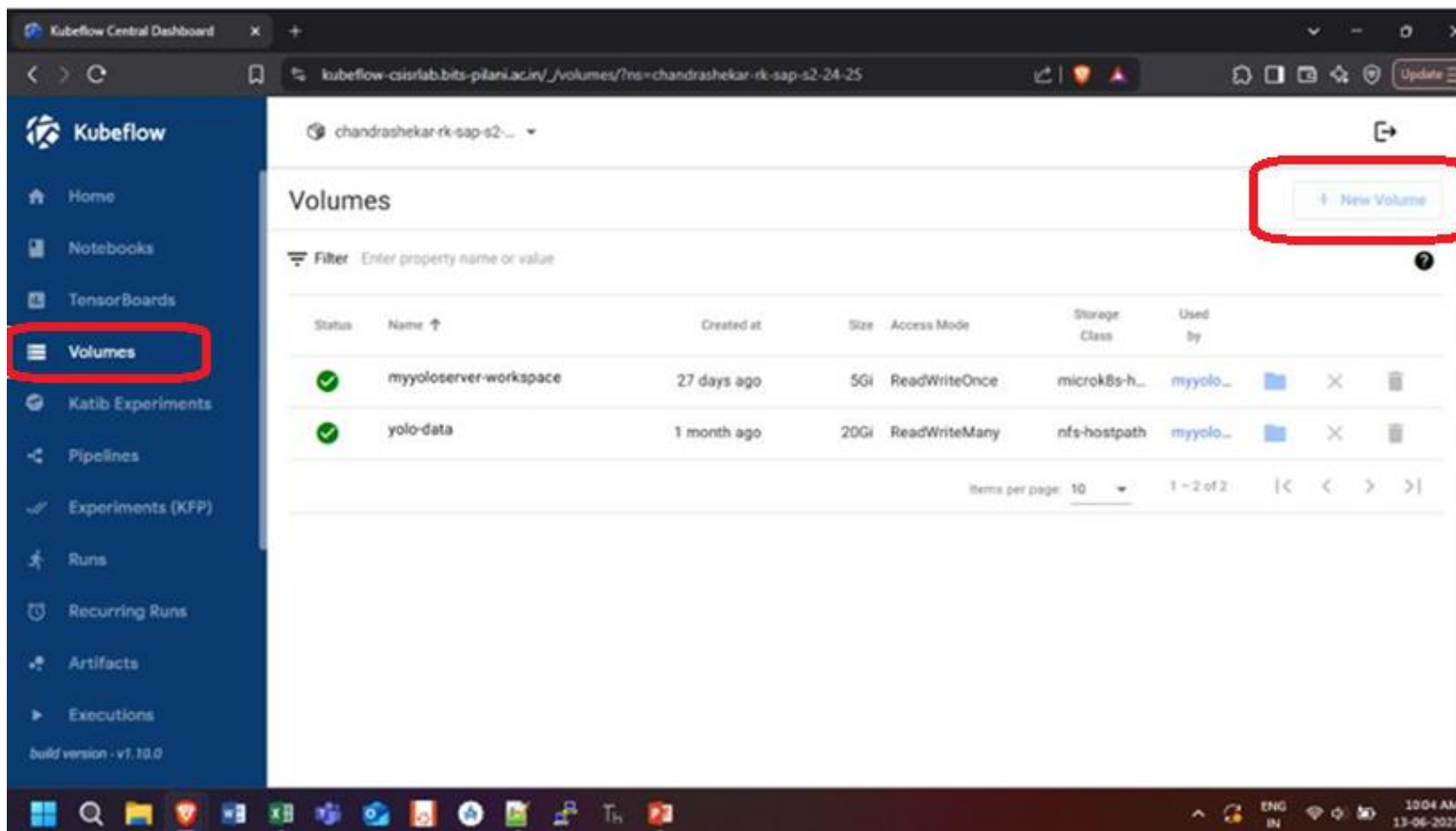
Designed and developed by [Code Argo](#)

innovate **achieve** **lead**

The Kubeflow portal landing page allows you to create notebooks for running your experiments.



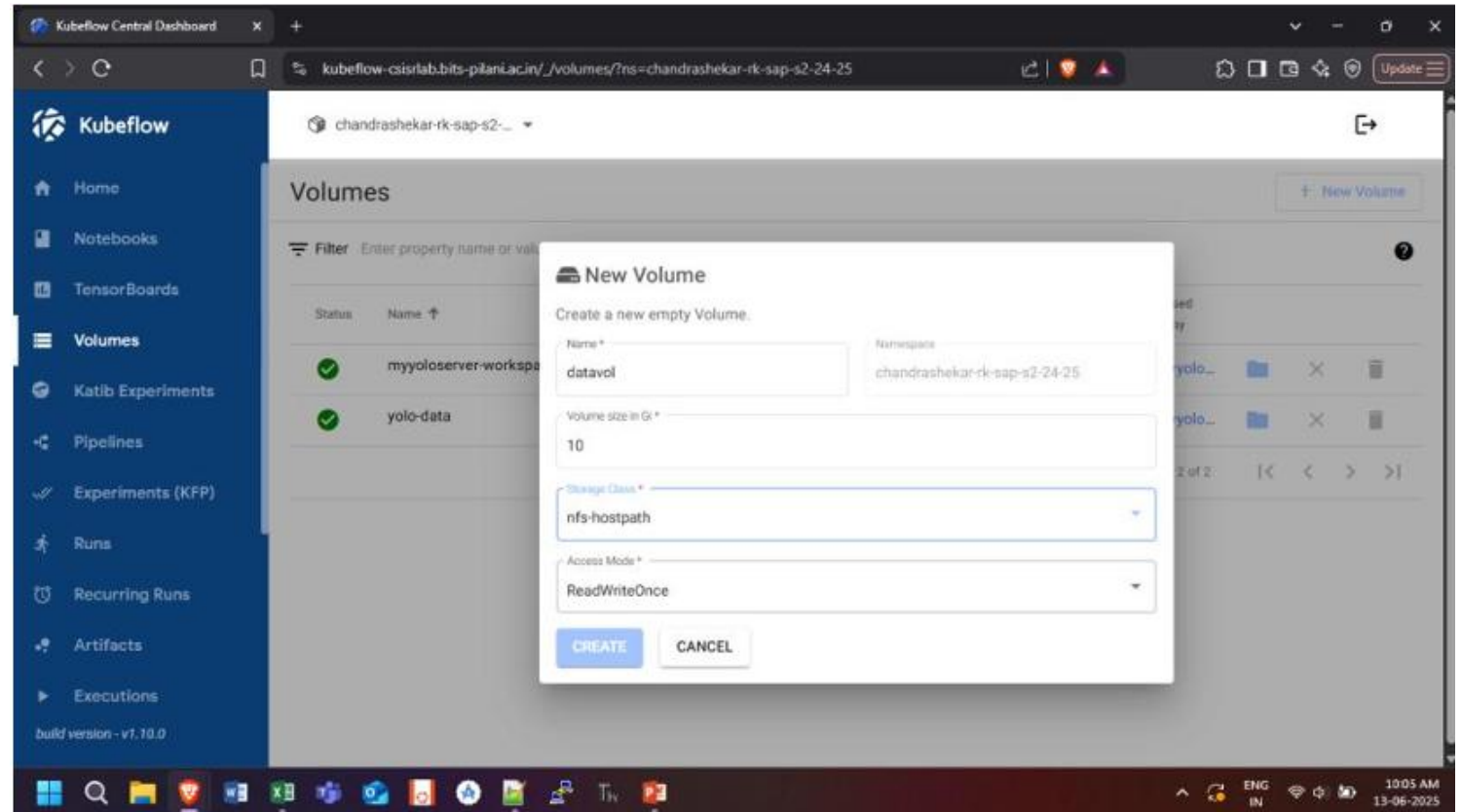
Step 2a. Create Data volume to store your data and source code in a NFS server.



Creating a New Data Volume

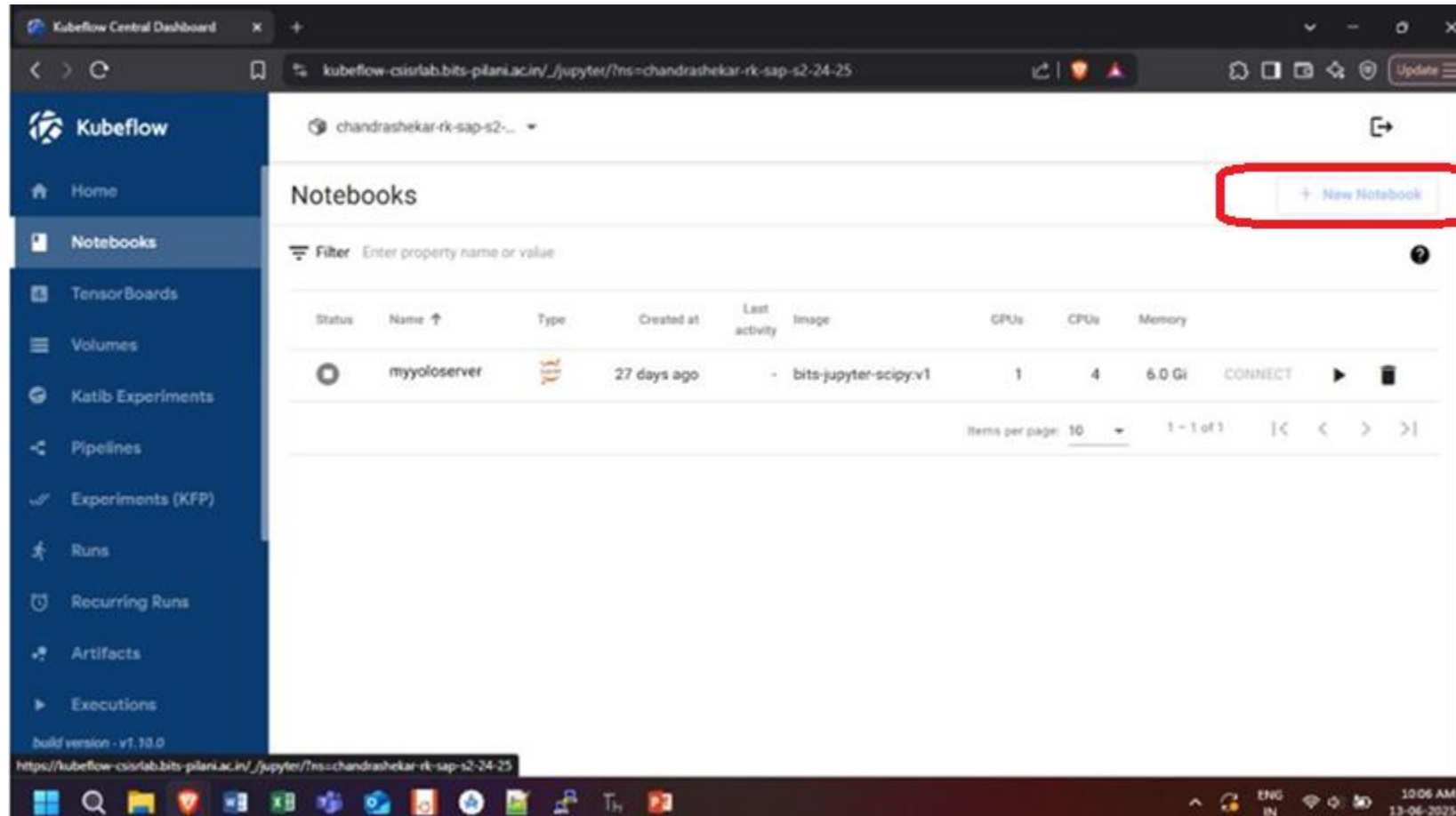
- Click on "+ New Volume" (if visible)
- **Name** your volume
- **Volume Size:** Keep the default 10 GB unless more is required (max 30 GB)
- **Storage Class:** Select **nfs-hostpath**
- **Access Mode:** Leave as default ReadWriteOnce
- Click "Create"

Note: Volume will get created only after being attached to a Notebook server



Step 2b: Create Notebook server

Click on “+ New Notebook”



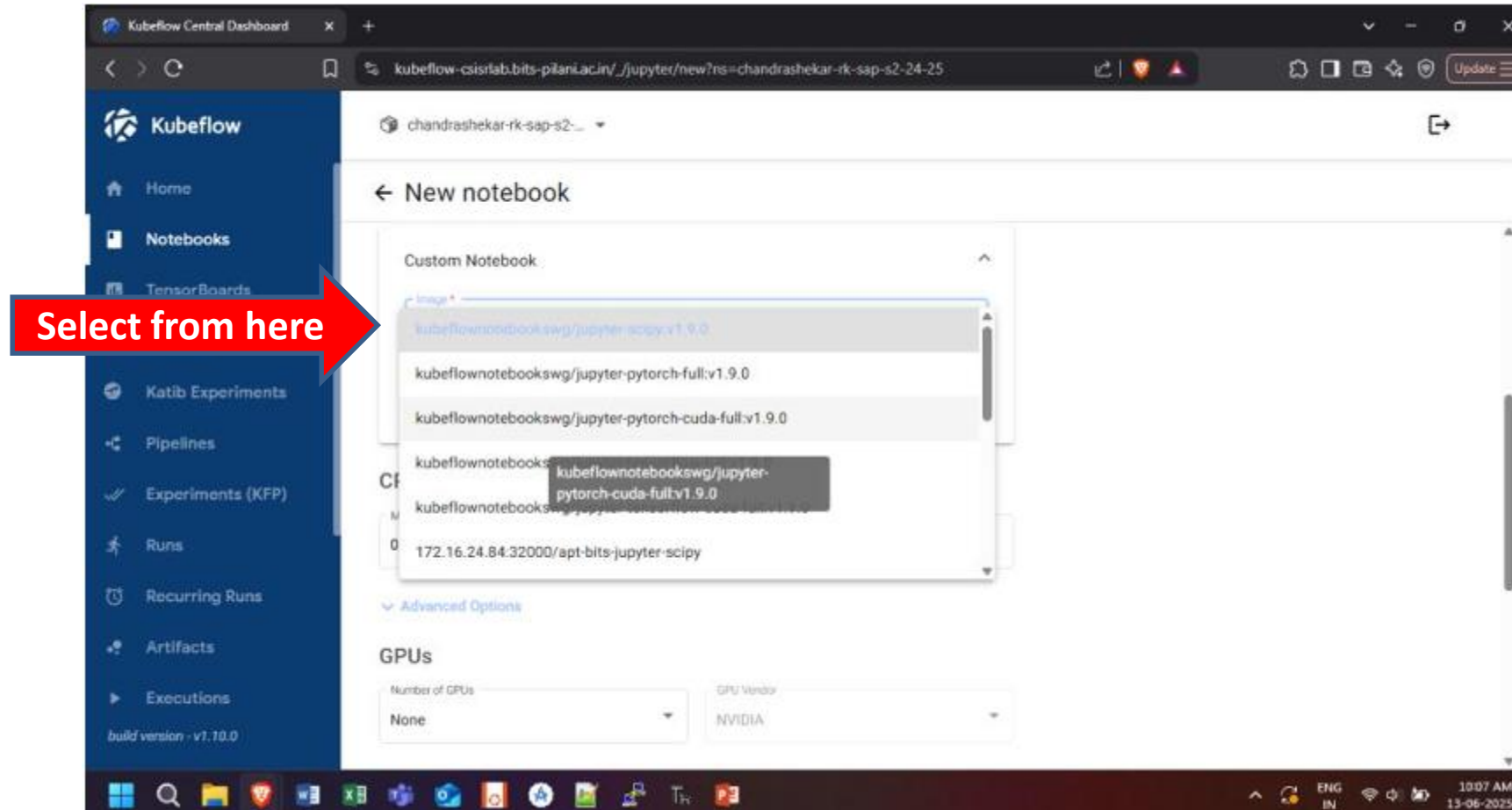
Enter a “Name” and Select “JupyterLab”

The screenshot shows the Kubeflow Central Dashboard interface. On the left is a dark blue sidebar with navigation links: Home, Notebooks, Volumes, Katib Experiments, Pipelines, Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled 'New notebook' and features a 'Name' input field with the text 'demo'. Below this is a grid of three options: JupyterLab (highlighted with a red arrow), VisualStudio Code (labeled '1'), and RStudio (labeled '2'). Each option includes a brief description. At the bottom of the grid is a 'Custom Notebook' button. The bottom of the dashboard shows 'CPU / RAM' information. The browser's address bar shows the URL 'kubeflow-csislab.bits-pilani.ac.in/_jupyter/new?ns=chandrashekar-rk-sap-s2-24-25'. The Windows taskbar at the very bottom shows the time as 10:06 AM on 13-06-2025.

Enter the name

Select JupyterLab

Select your preferred “Custom Notebook” from the dropdown.



CPU/RAM

- Enter Minimum(0.5)/Maximum(4) CPU cores. Enter same value for both Min and Max
- Enter Minimum (1)/ Maximum(16) Memory(RAM) in GB. Enter same values for both Min and Max

The screenshot shows the 'New notebook' page in the Kubeflow Central Dashboard. The left sidebar contains navigation links: Home, Notebooks, TensorBoards, Volumes, Katib Experiments, Pipelines, Experiments (KFP), Recurring Runs, and Executions. The main content area is titled 'New notebook' and includes a 'Custom Notebook' section with an 'Image' dropdown set to '172.16.24.84:32000/apt-bits-jupyter-scipy'. Below this is the 'CPU / RAM' section, which contains four input fields: 'Minimum CPU *' (value: 2), 'Minimum Memory Gi *' (value: 4), 'Maximum CPU' (value: 2), and 'Maximum Memory Gi' (value: 4). Two red arrows with white text point to these fields: 'Enter CPU' points to the 'Minimum CPU' field, and 'Enter Memory' points to the 'Maximum Memory Gi' field. Below the CPU/RAM section is an 'Advanced Options' link and a 'GPUs' section. The bottom of the image shows a Windows taskbar with various application icons and a system tray showing the time as 10:08 AM on 13-06-2025.

Kubeflow Central Dashboard

kubeflow-csisrlab.bits-pilani.ac.in/_jupyter/new?ns=chandrashekar-rk-sap-s2-24-25

chandrashekar-rk-sap-s2-...

← New notebook

Custom Notebook

Image *

172.16.24.84:32000/apt-bits-jupyter-scipy

Advanced Options

CPU / RAM ?

Minimum CPU *

2

Minimum Memory Gi *

4

Maximum CPU

2

Maximum Memory Gi

4

Advanced Options

GPUs

Number of GPUs

GPUs

Enter CPU

Enter Memory

Executions

build version - v1.10.0

10:08 AM 13-06-2025

Select GPU – 1 (Max allowed)

The screenshot shows the Kubeflow Central Dashboard interface. On the left is a dark blue sidebar with navigation links: Home, Notebooks, TensorBoards, Volumes, Pipelines, Experiments (KFP), Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled 'New notebook' and includes sections for 'GPUs' and 'Data Volumes'. In the 'GPUs' section, the 'Number of GPUs' dropdown menu is open, showing options: None, 1, 2, 4, and 8. A red arrow points from the sidebar to this dropdown with the text 'Select GPU'. To the right, the 'GPU Vendor' dropdown is set to 'NVIDIA', with another red arrow pointing to it from the right with the text 'Select NVIDIA'. Below these are fields for 'Storage' and 'Space, Empty, 5Gi'. The 'Data Volumes' section has buttons for '+ Add new volume' and '+ Attach existing volume'. The bottom of the image shows a Windows taskbar with various application icons and a system tray displaying '10:08 AM' and '13-06-2025'.

Workspace Volume

Retain the default name and storage space 5GB (if you are installing large packages for training purpose, you may increase the storage space)

Kubeflow Central Dashboard

kubeflow-csisrlab.bits-pilani.ac.in/_jupyter/new?ns=chandrashekar-rk-sap-s2-24-25

chandrashekar-rk-sap-s2-...

← New notebook

Data Volumes

Additional volumes that will be mounted in your Notebook.

Existing volume

Type: Kubernetes Volume

☐ Readonly

Name	Storage	Access Mode
datavol	10Gi	ReadWriteOnce
myyolosever-workspace	5Gi	ReadWriteOnce
yolo-data	20Gi	ReadWriteMany

+ Add new volume + Attach existing volume

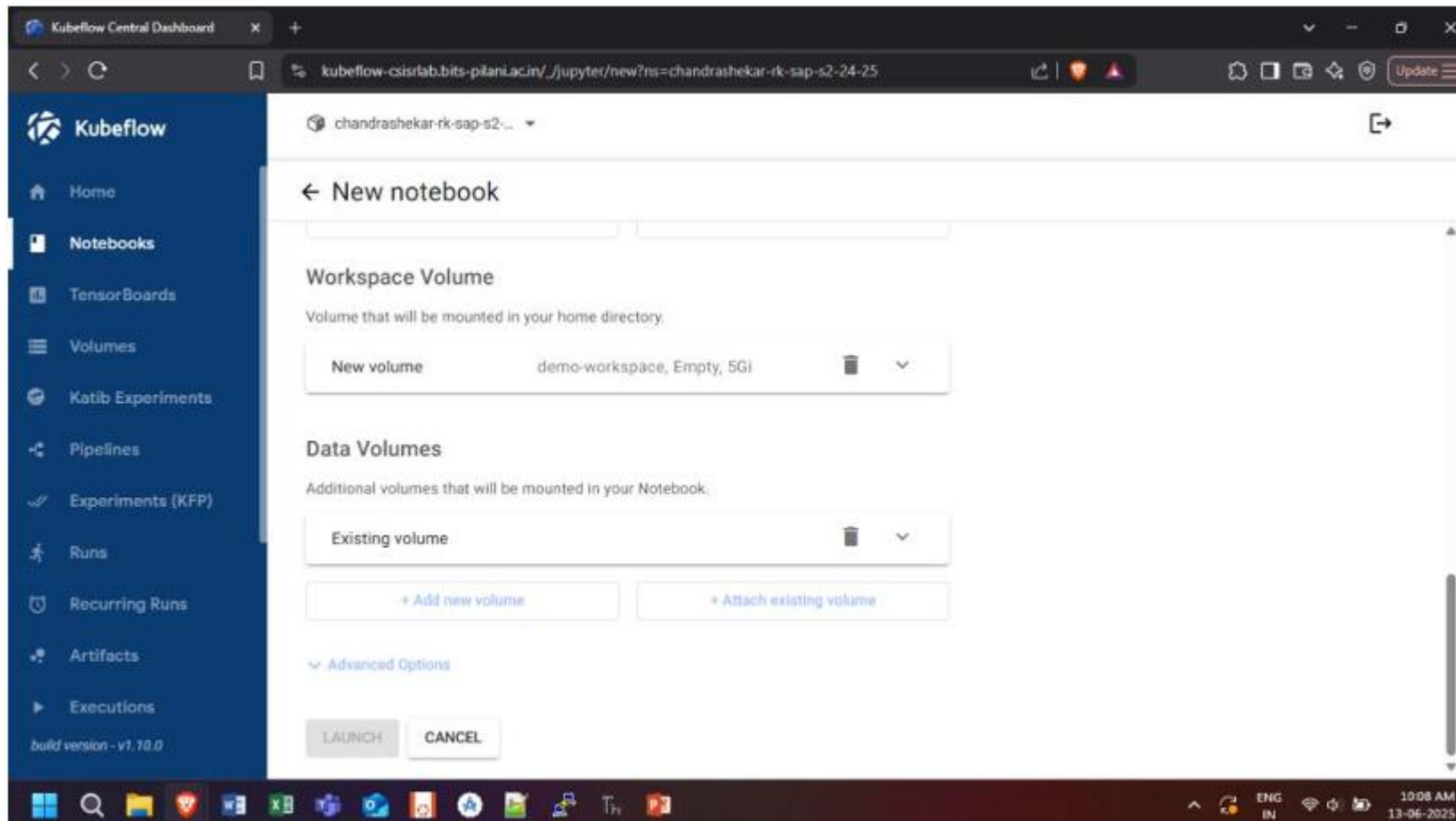
Advanced Options

build version - v1.10.0

10:09 AM 13-06-2025

Data Volume:

Click on the dropdown for “Existing Volumes” and select the Data Volume which you have created in Step 2a.

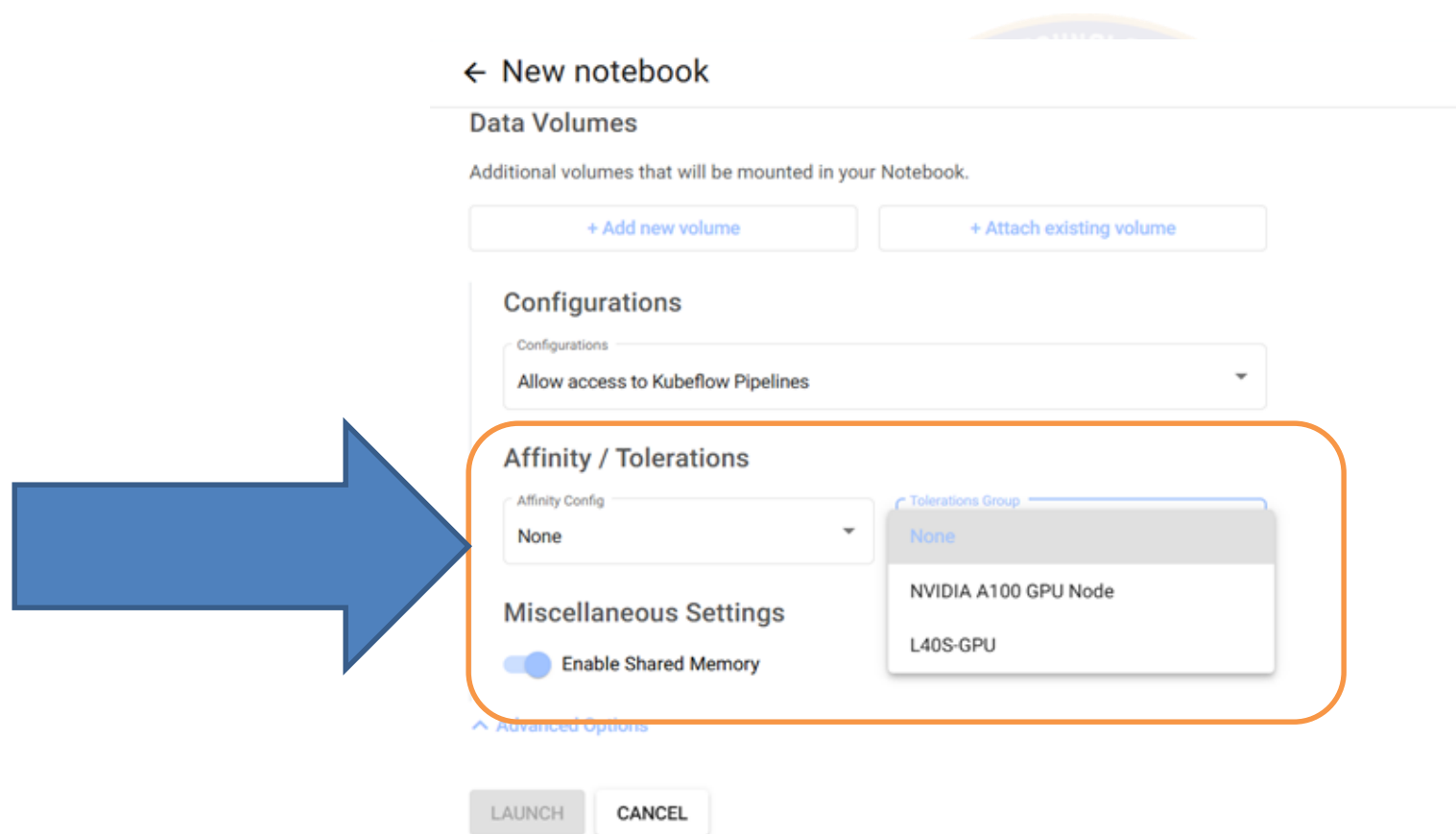


GPU selection:

Click on the “Advanced Options” to select the GPU.

Affinity /Tolerations → Affinity config – None

Toleration Group → Select GPU



← New notebook

Data Volumes

Additional volumes that will be mounted in your Notebook.

+ Add new volume + Attach existing volume

Configurations

Configurations
Allow access to Kubeflow Pipelines

Affinity / Tolerations

Affinity Config
None

Miscellaneous Settings

Enable Shared Memory

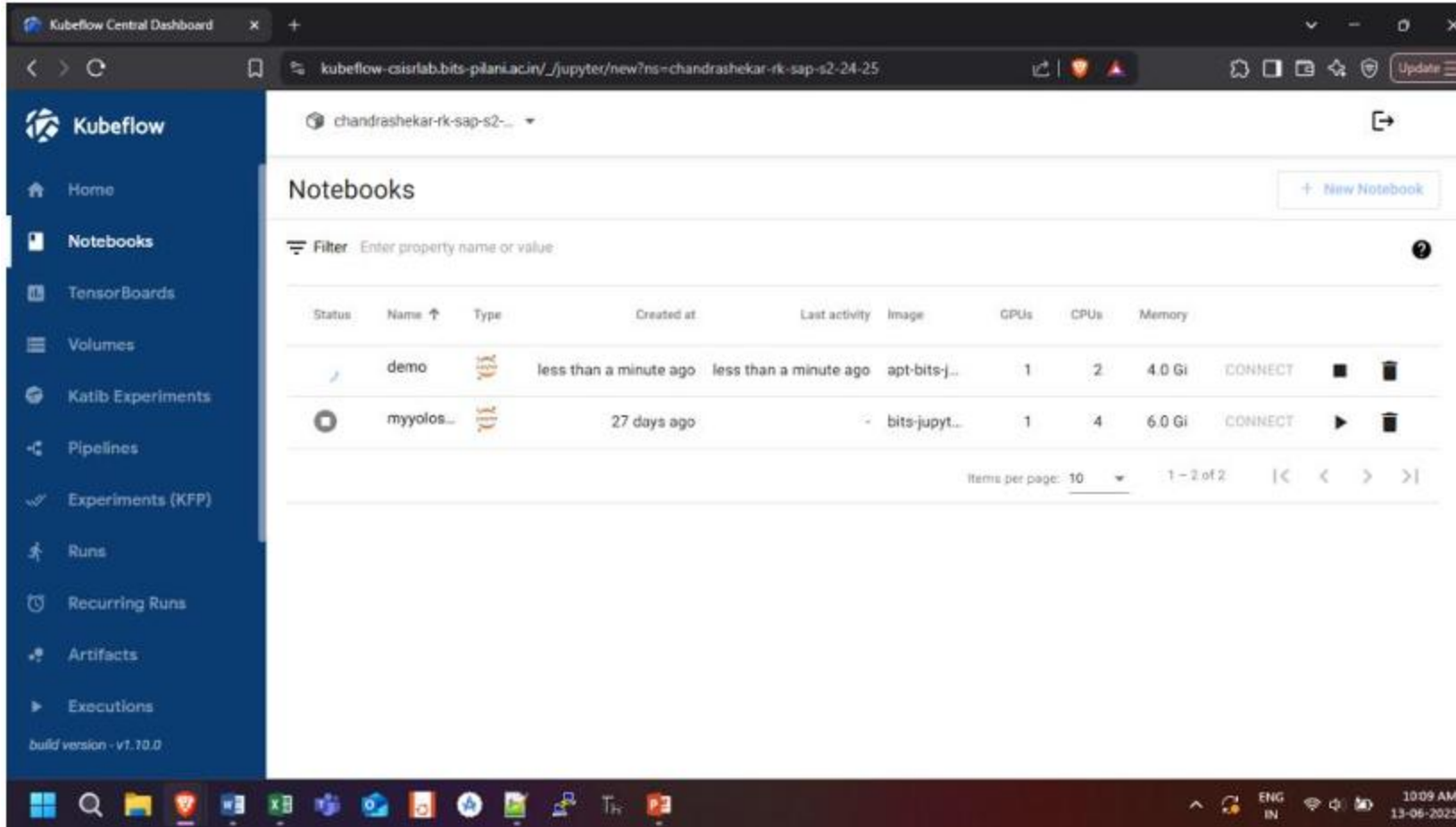
Tolerations Group

- None
- NVIDIA A100 GPU Node
- L40S-GPU

Advanced Options

LAUNCH CANCEL

Go back to Notebooks to see your Notebook server getting launched.



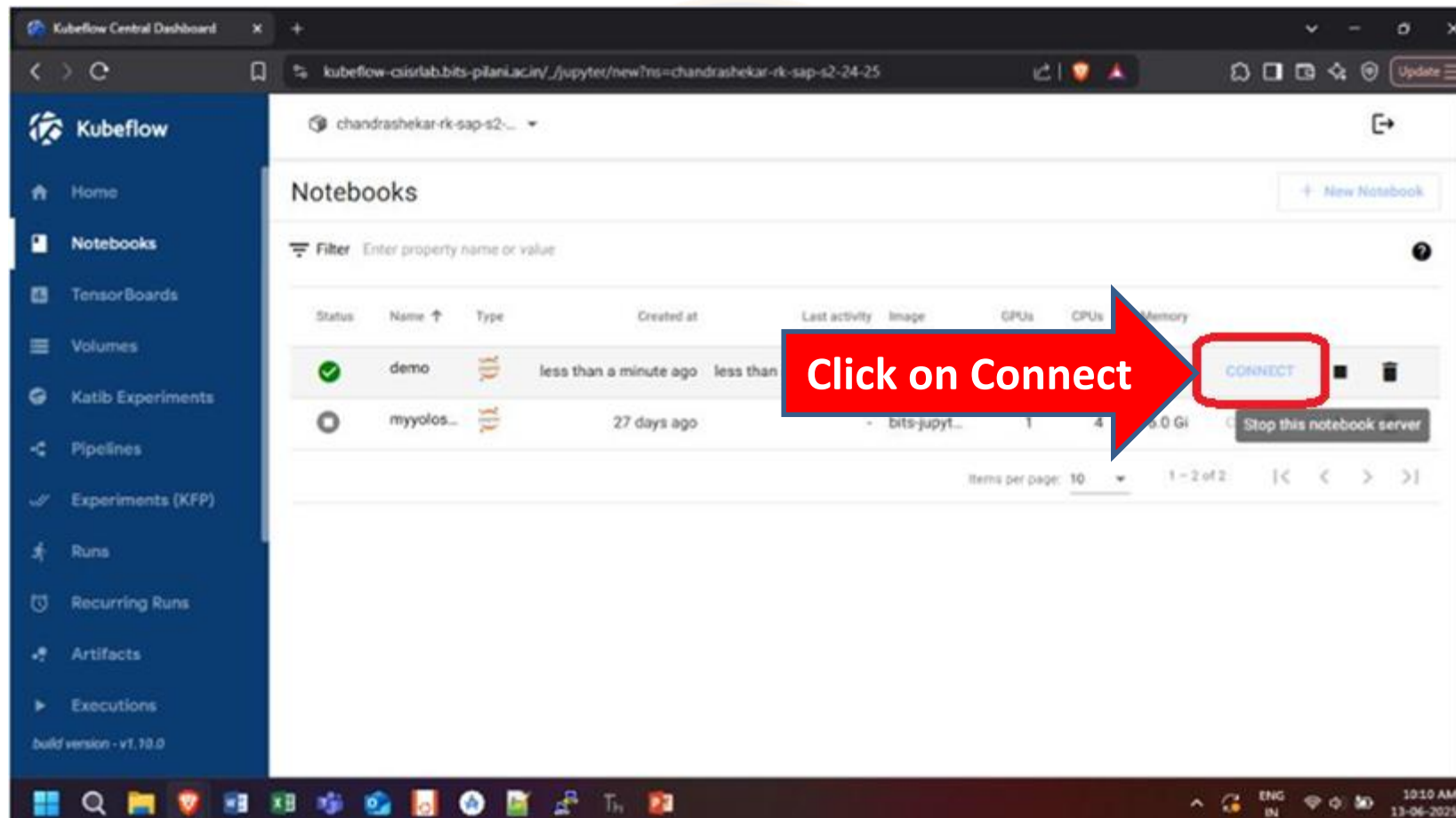
The screenshot displays the Kubeflow Central Dashboard in a web browser. The left sidebar contains navigation links: Home, Notebooks (selected), TensorBoards, Volumes, Katib Experiments, Pipelines, Experiments (KFP), Runs, Recurring Runs, Artifacts, and Executions. The main content area is titled 'Notebooks' and includes a '+ New Notebook' button. Below this is a filter input field. A table lists the notebooks:

Status	Name	Type	Created at	Last activity	Image	GPUs	CPUs	Memory	Actions
	demo		less than a minute ago	less than a minute ago	apt-bits-j...	1	2	4.0 Gi	CONNECT
	myyolos...		27 days ago		bits-jupyt...	1	4	6.0 Gi	CONNECT

At the bottom right of the table, it shows 'Items per page: 10' and '1 - 2 of 2'. The Windows taskbar at the bottom indicates the time is 10:09 AM on 13-06-2025.

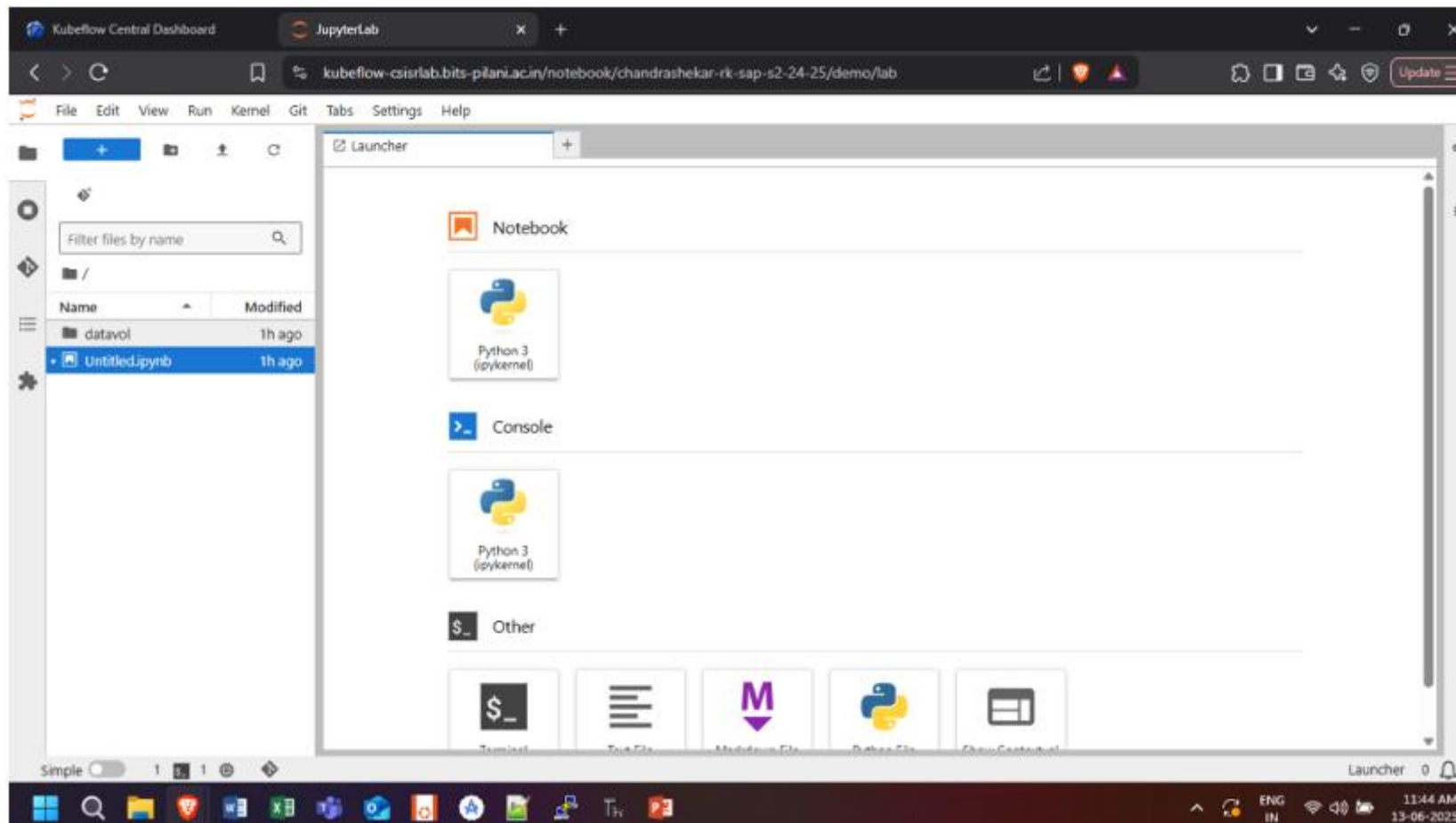
Click on “Connect” button to connect to the Notebook server. Once you have finished your work, click on “STOP” icon.

If you need to delete your Notebook server, click on the “Delete” icon

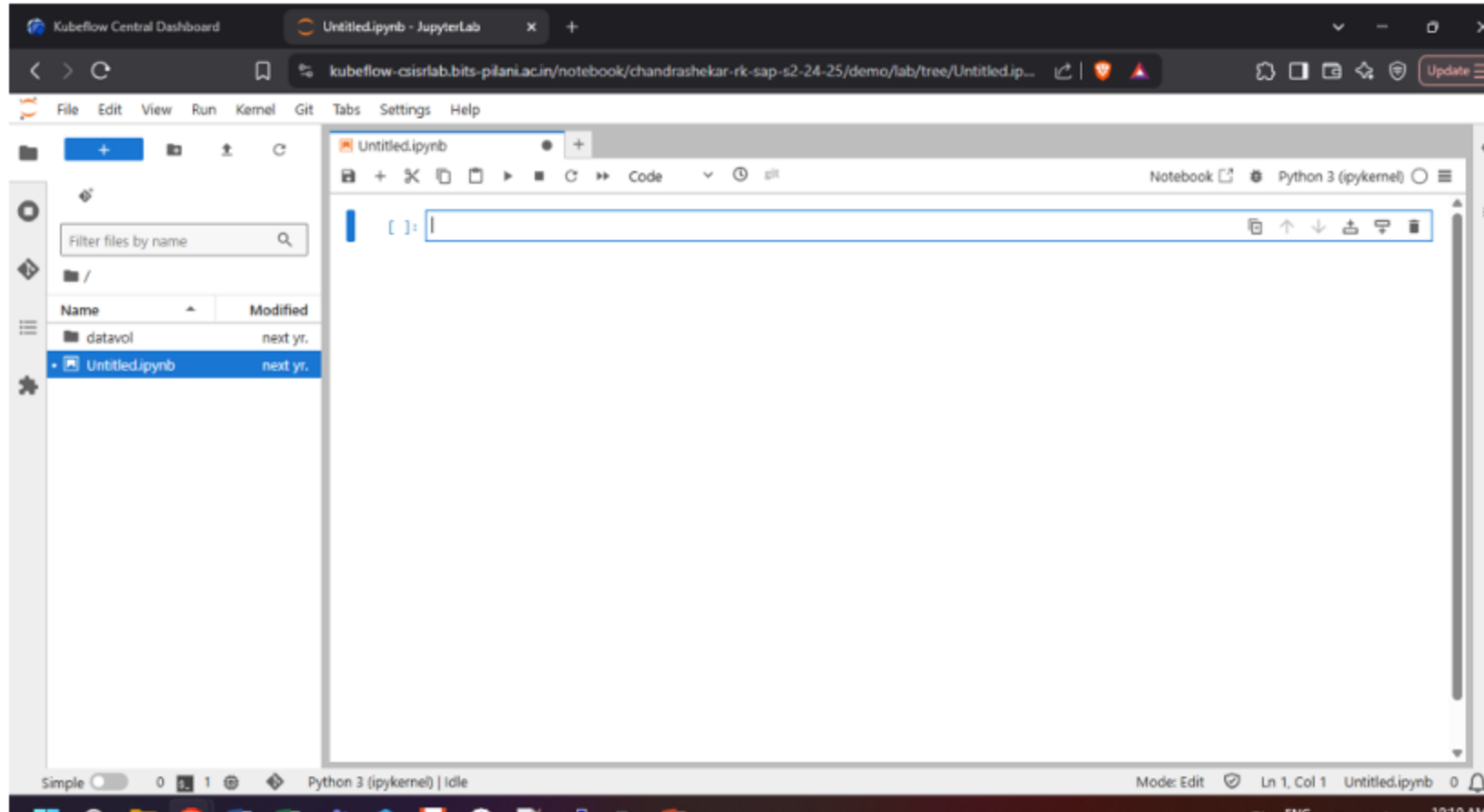


Notebook server interface which can launch – Notebook, Terminal etc.

Left-hand side, you will see the Data Volume attached as a folder to this Notebook server



This is the Notebook interface



You are now ready to use the GPU cluster efficiently. Always manage resources wisely and backup your work!

Notebook interface[Contd..]

You may run Ubuntu(Linux) commands here.

- Create virtual environment
- Install packages using pip install
- Zip your files or folders to download to your local machine

Recursively Zip a Directory into one archive

```
zip -r [file_name.zip] [directory_name]
```

- Unzip
unzip [file_name.zip]



Technical Support details

Technical SPOC: **Mr. Bangaru Santosh Kumar**
Mail id: **csislabsupport@wilp.bits-pilani.ac.in**

Thank You

